

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SAAWK")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6665, C3 = 1214)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,          TruncNormal,              0.1,          0.2,          0.01, 0.8);
Distrib (r_pAm_pM,    TruncNormal_cv,          0.8,          0.1,          0.1, 2);
Distrib (kap,         Uniform,                  0.1,          0.9);
Distrib (kapA,        Uniform,                  0.1,          0.9);
Distrib (Ehp,         TruncNormal_cv,          100,          0.2,          40, 300);
Distrib (E_coc,       TruncNormal_cv,          30,          0.2,          1, 300);
Distrib (rOM_ClXHorse, LogUniform,             0.00001,      0.3);
Distrib (K, Uniform, 0.00001, 2);
Distrib (dv, TruncNormal_cv, 2, 0.5, 0, 10);
Distrib (w0, LogUniform, 0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcaliAWK DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	kapA.1.	Ehp.1.
Result_mode	110.08500	0.78617900	0.96002300	0.899748000	0.59512200	188.39400
2.5%	77.63090	0.64965200	0.81864400	0.890339150	0.45451700	160.01015
97.5%	130.19300	0.79877900	1.09220700	0.899936000	0.67682030	211.38580

Min	70.09200	0.04100250	0.73469400	0.273930000	0.14752400	63.11510
Max	146.77400	0.79998800	1.23138000	0.899999000	0.72357100	242.25100
Result_mean	106.08789	0.75332519	0.95604272	0.897391844	0.57690012	186.14196
Result_sd	13.65179	0.04058884	0.06919178	0.005417572	0.05503437	13.26627
	E_coc.1.	rOM_ClxFHorse.1.	K.1.	dv.1.	w0.1.	
Result_mode	35.74250	0.0037356400	0.43412100	2.216370	2.293000e-09	
2.5%	30.83230	0.0023873700	0.34135300	1.832589	1.619760e-09	
97.5%	43.69860	0.0056776300	0.56423900	2.603760	2.555521e-05	
Min	22.35270	0.0005039450	0.29201300	1.231370	1.242940e-09	
Max	49.51890	0.0077377100	1.03606000	3.247670	6.372390e-03	
Result_mean	37.20931	0.0039226515	0.44918697	2.197782	3.255672e-06	
Result_sd	3.28248	0.0008363363	0.05828205	0.201513	3.735728e-05	
	Sigma_W.1.					
Result_mode	0.11822600					
2.5%	0.10812900					
97.5%	0.13976600					
Min	0.09800740					
Max	5.13596000					
Result_mean	0.12333615					
Result_sd	0.04694786					

Number of observations, n = 277

Number of parameters, k = 11

Log-likelihood, LL = -10.27232

BIC = 82.40883

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.01	1.02
E_coc.1.	1.03	1.08
Ehp.1.	1.05	1.15
Fm.1.	1.00	1.00
K.1.	1.49	2.25
kap.1.	1.02	1.03
kapA.1.	1.13	1.37
pAm.1.	1.43	2.11
r_pAm_pM.1.	1.04	1.13
rOM_ClxFHorse.1.	1.01	1.04
Sigma_W.1.	1.00	1.00
w0.1.	1.08	1.21

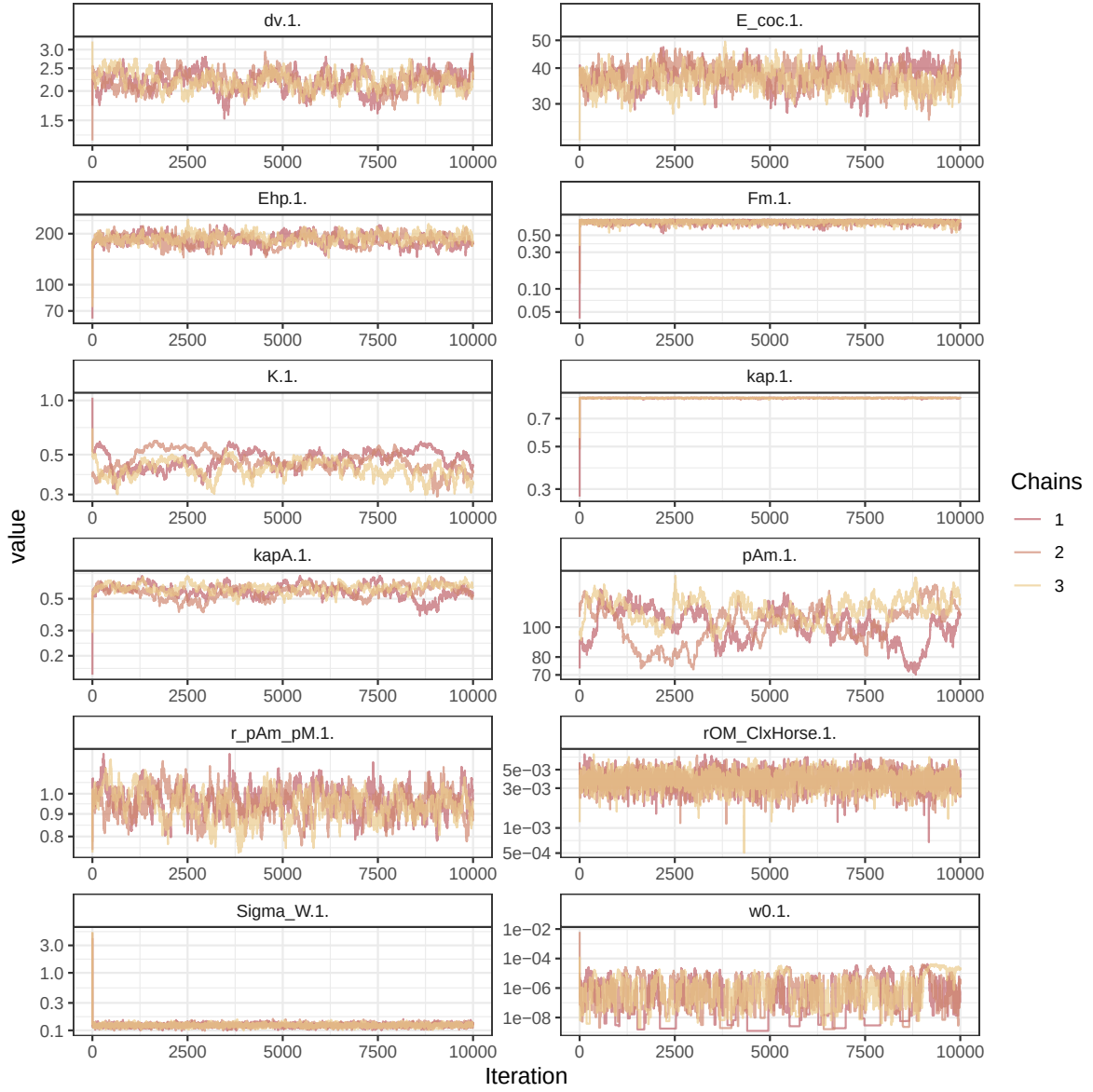


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

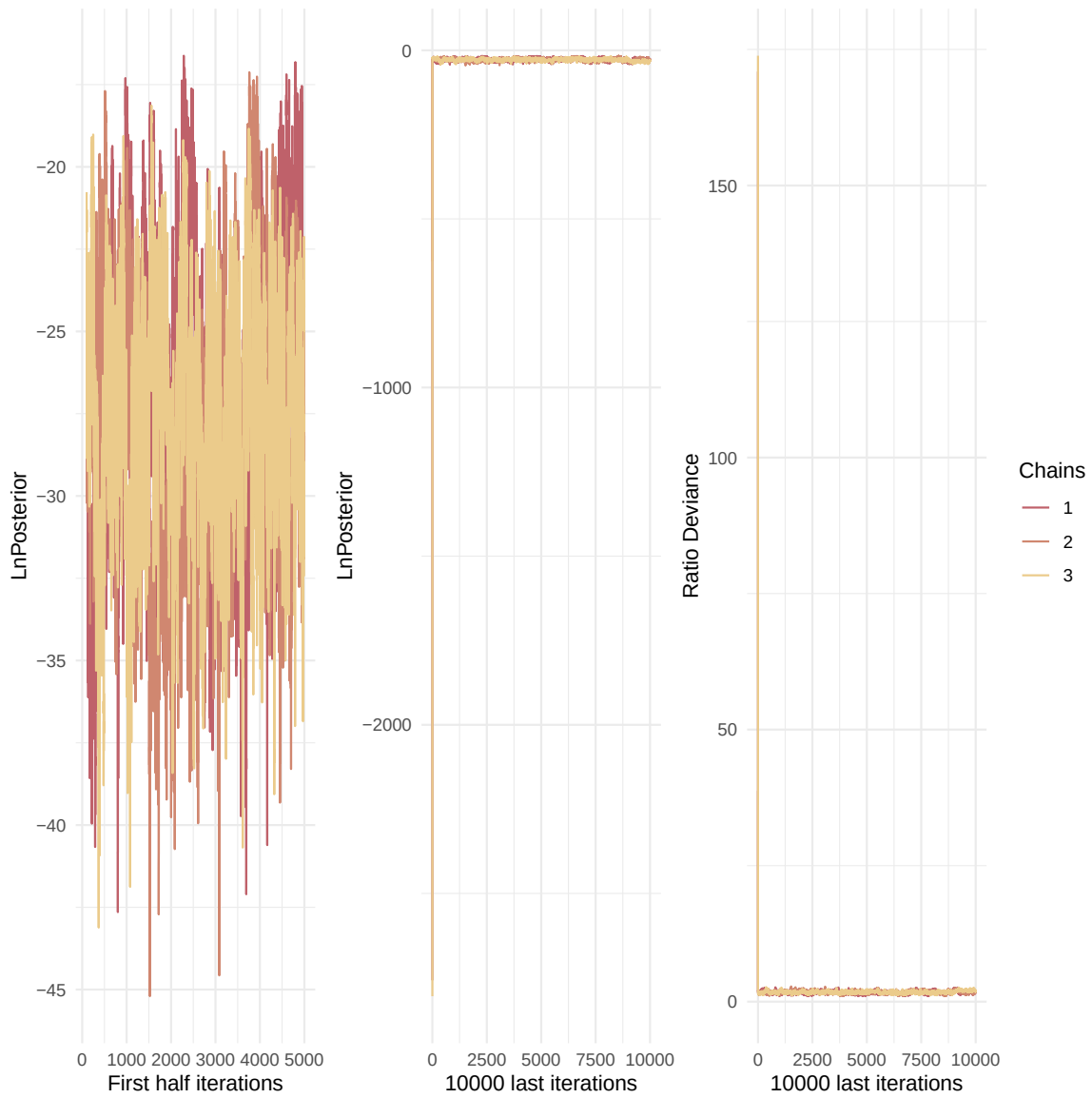


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

Multivariate psrf

1.35

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

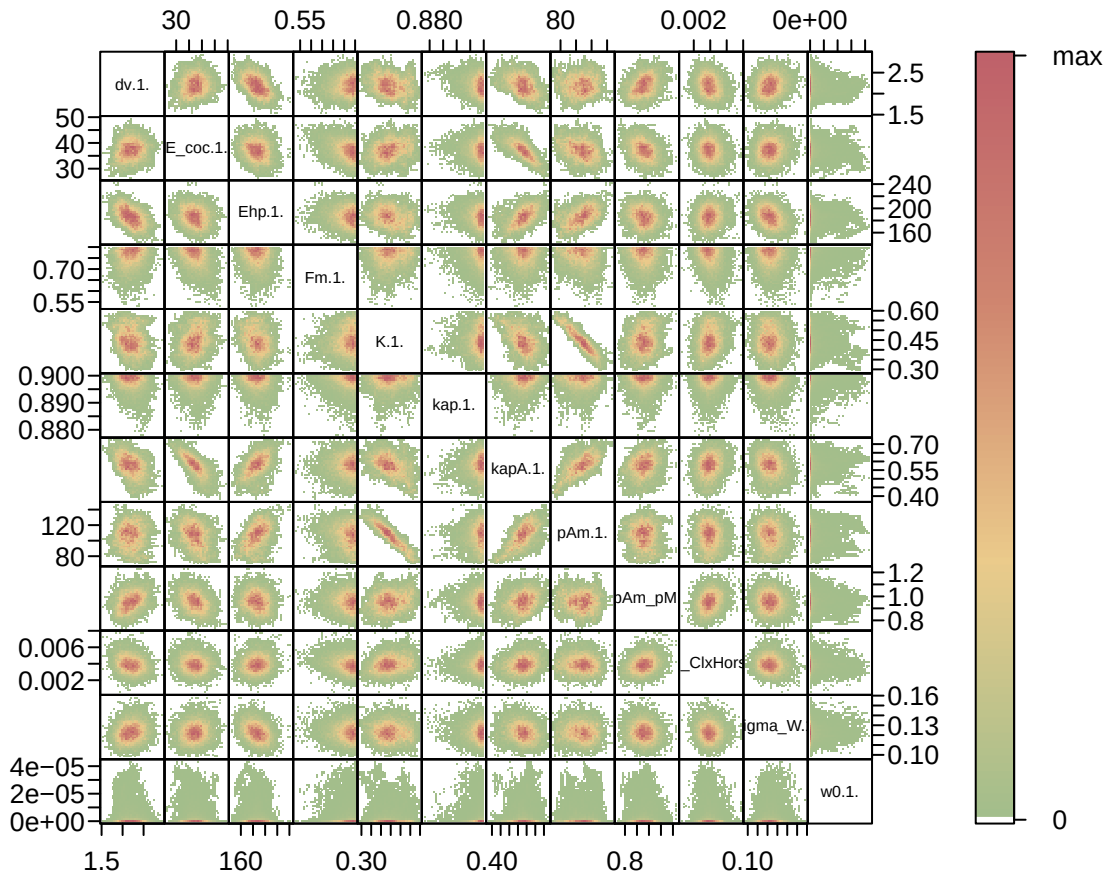


Figure 5: Parameter correlations.

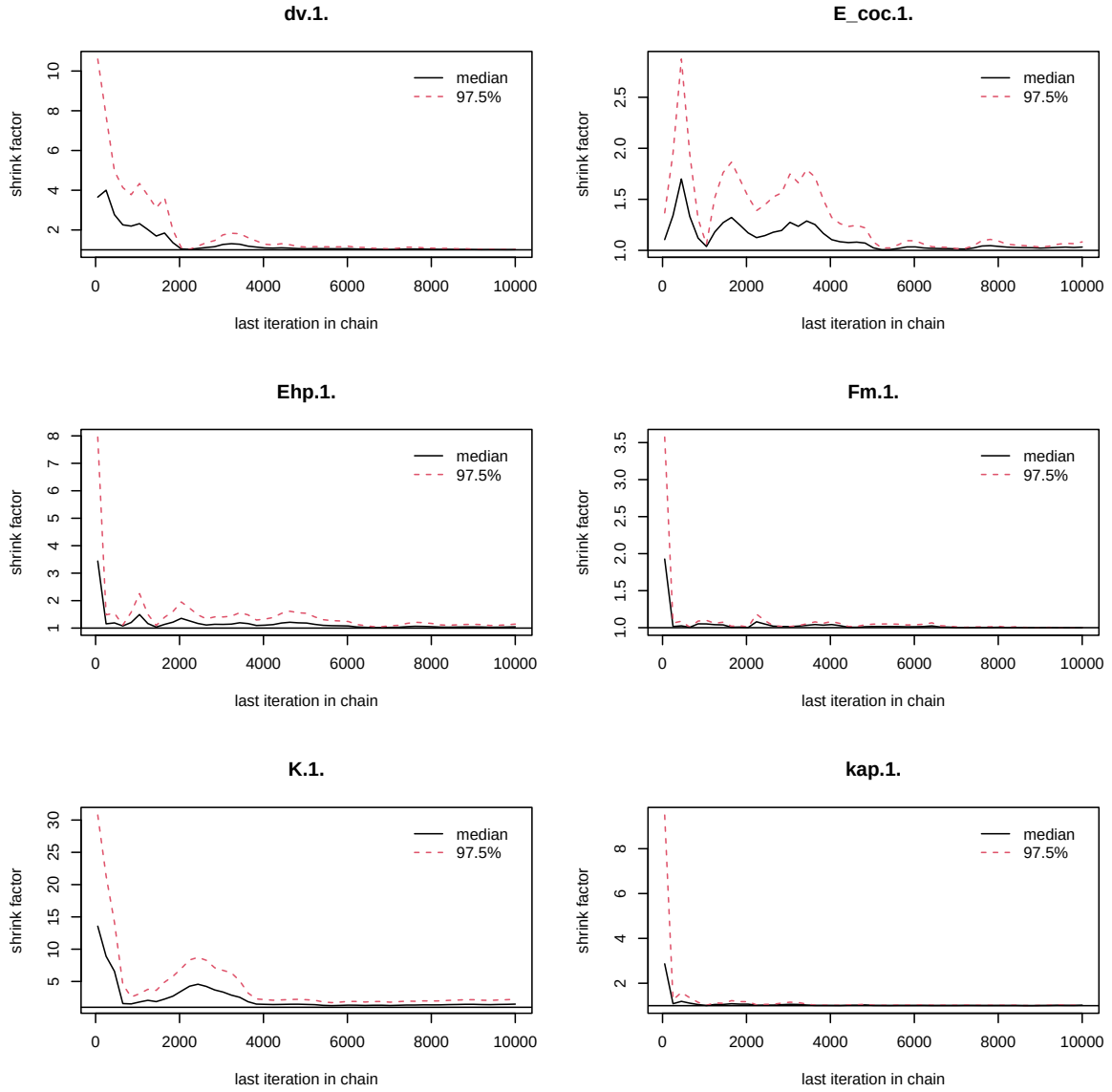


Figure 3: Chains convergence.

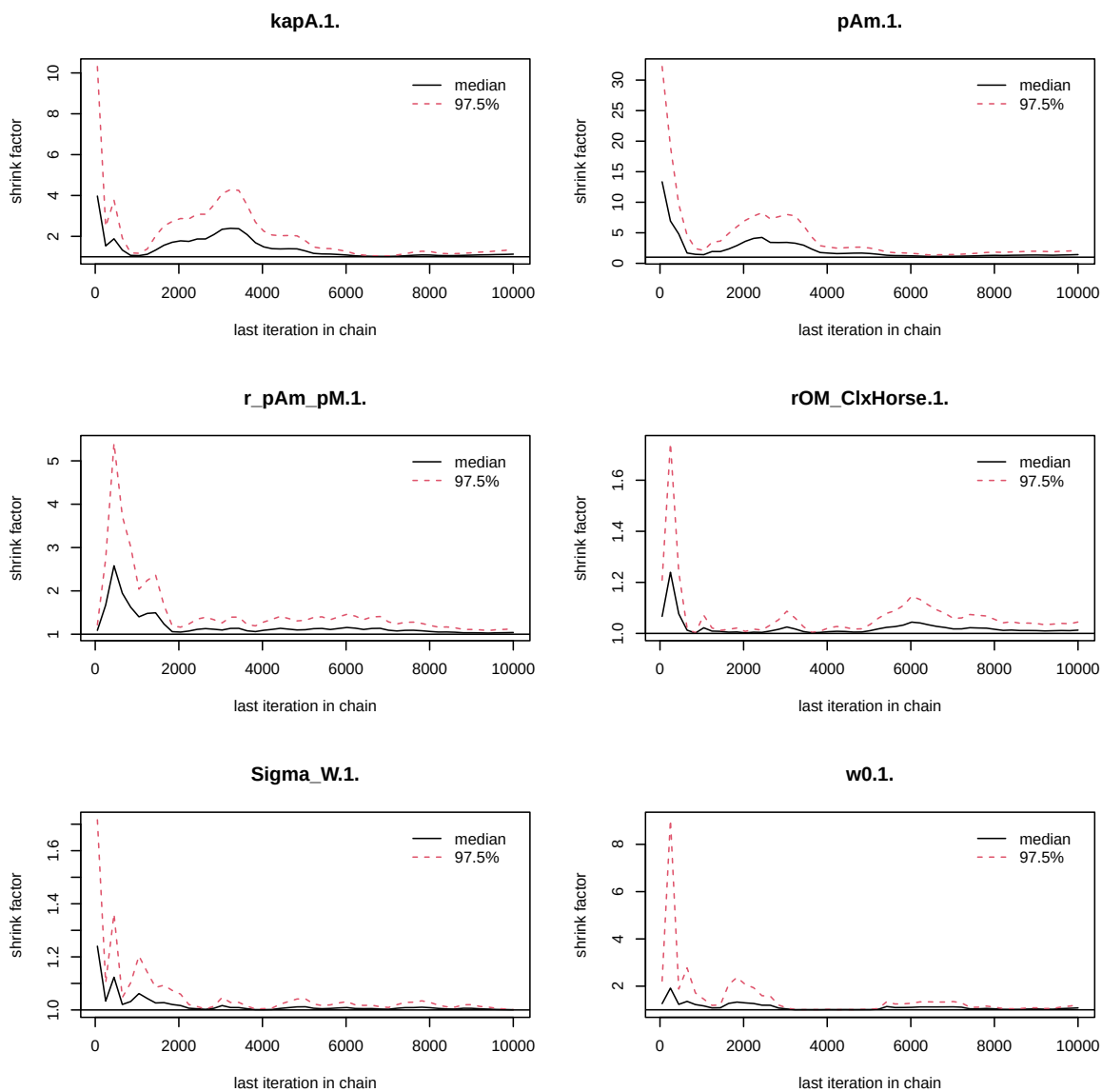


Figure 4: Chains convergence.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\\. $|\\.1\\\. ", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"
```

Posterior
and **prior** distributions

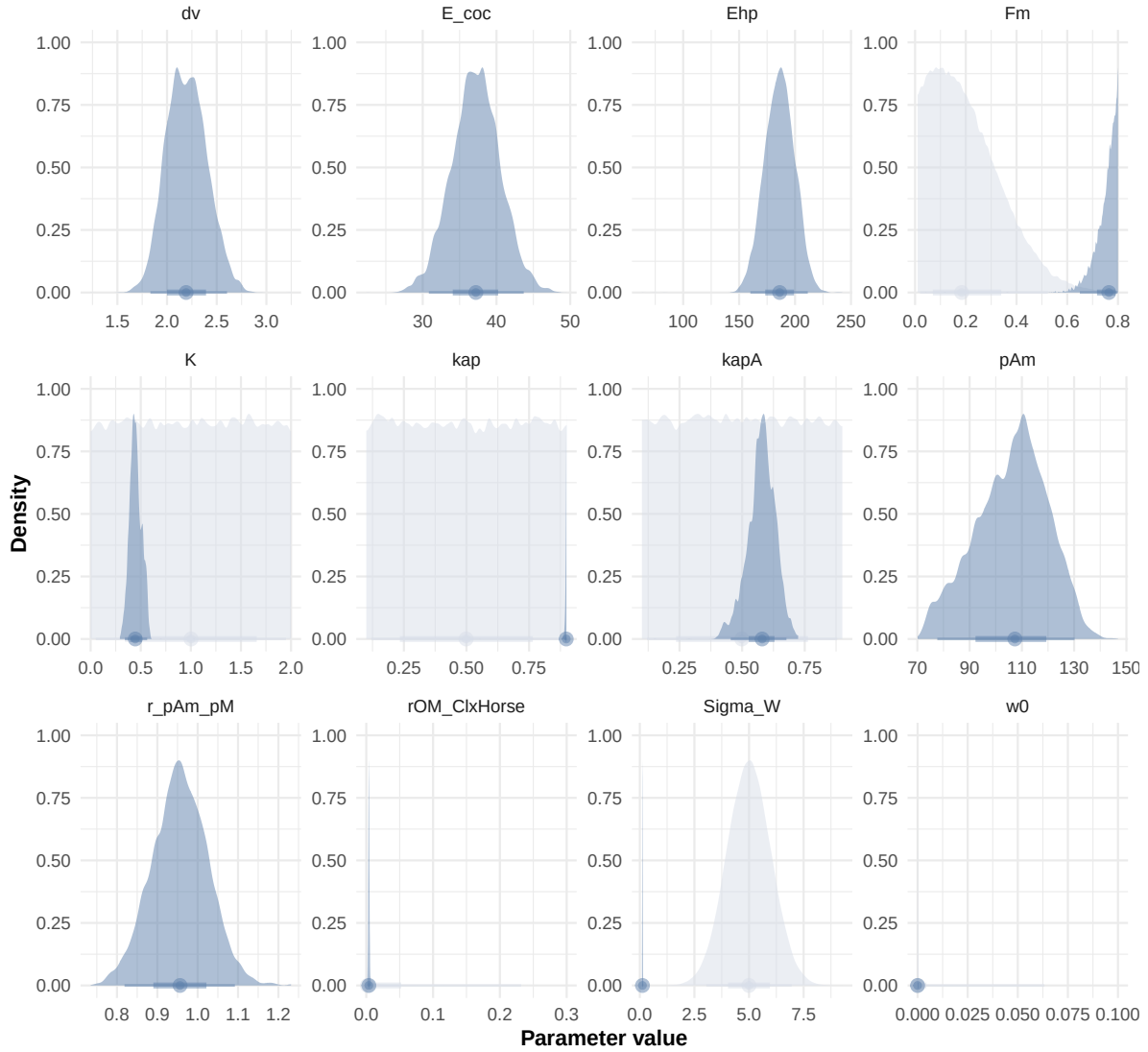


Figure 6: Distributions of the priors and the posteriors.

```
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_print)

# ./mcsim.DEBcaliAWK DEB_sim_full.in
```

```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

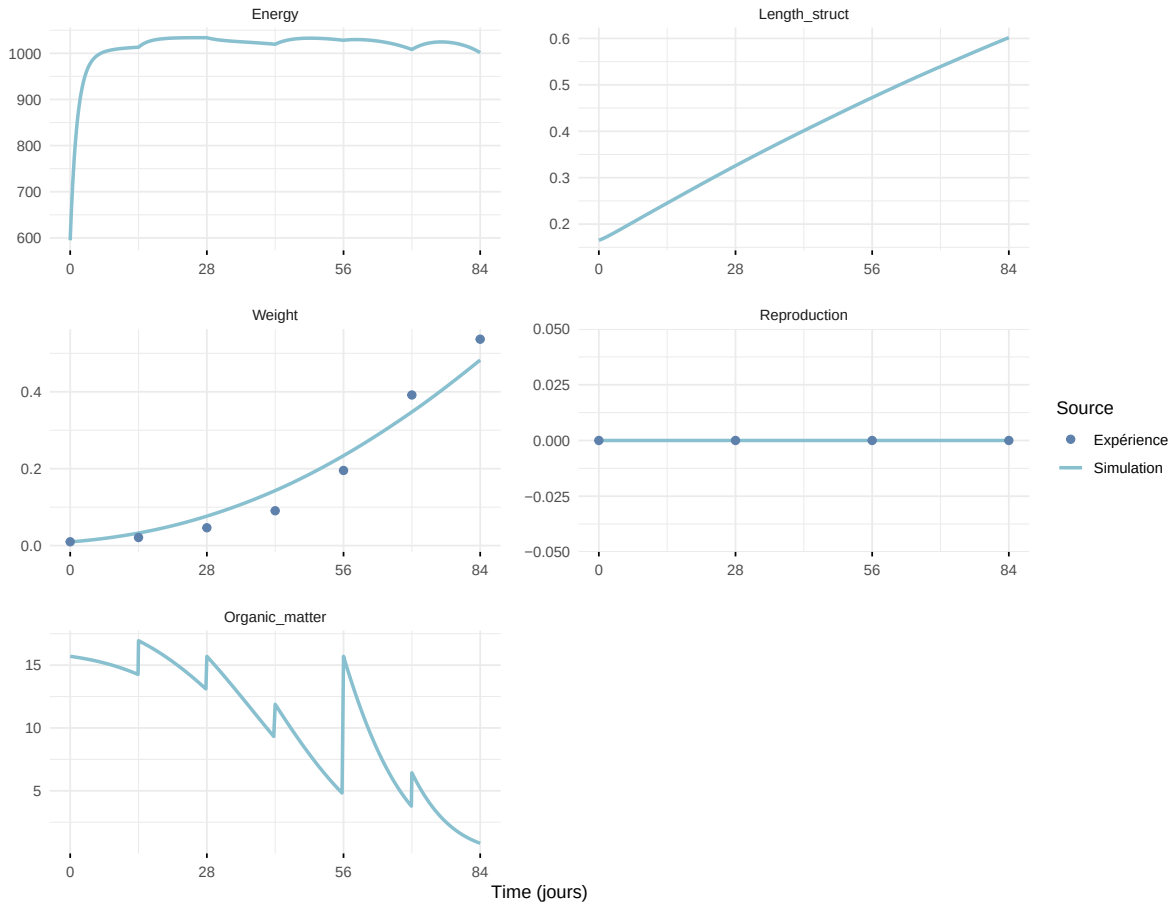
select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexp
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

Expérience : Bart2019_XP1_1



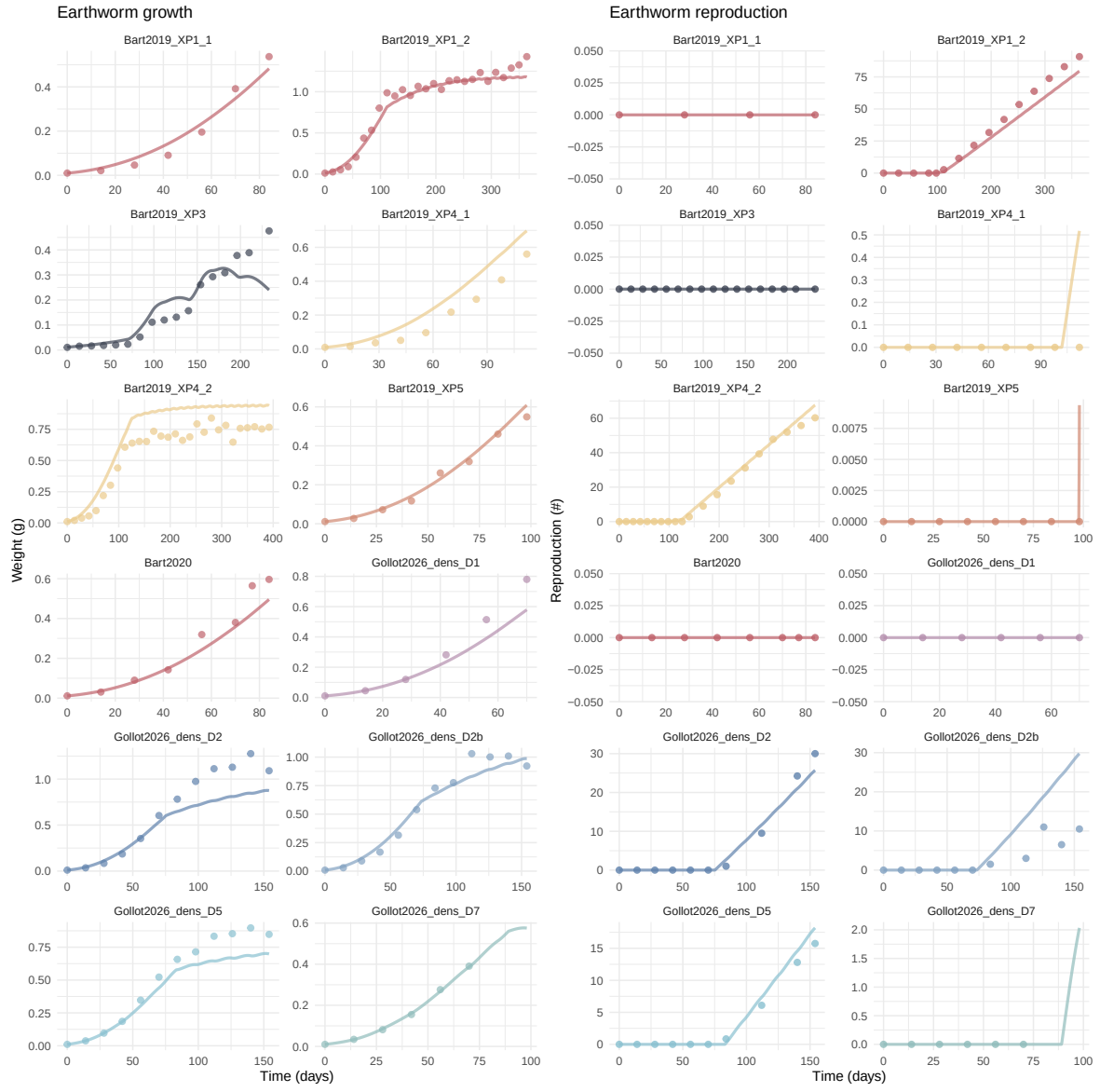
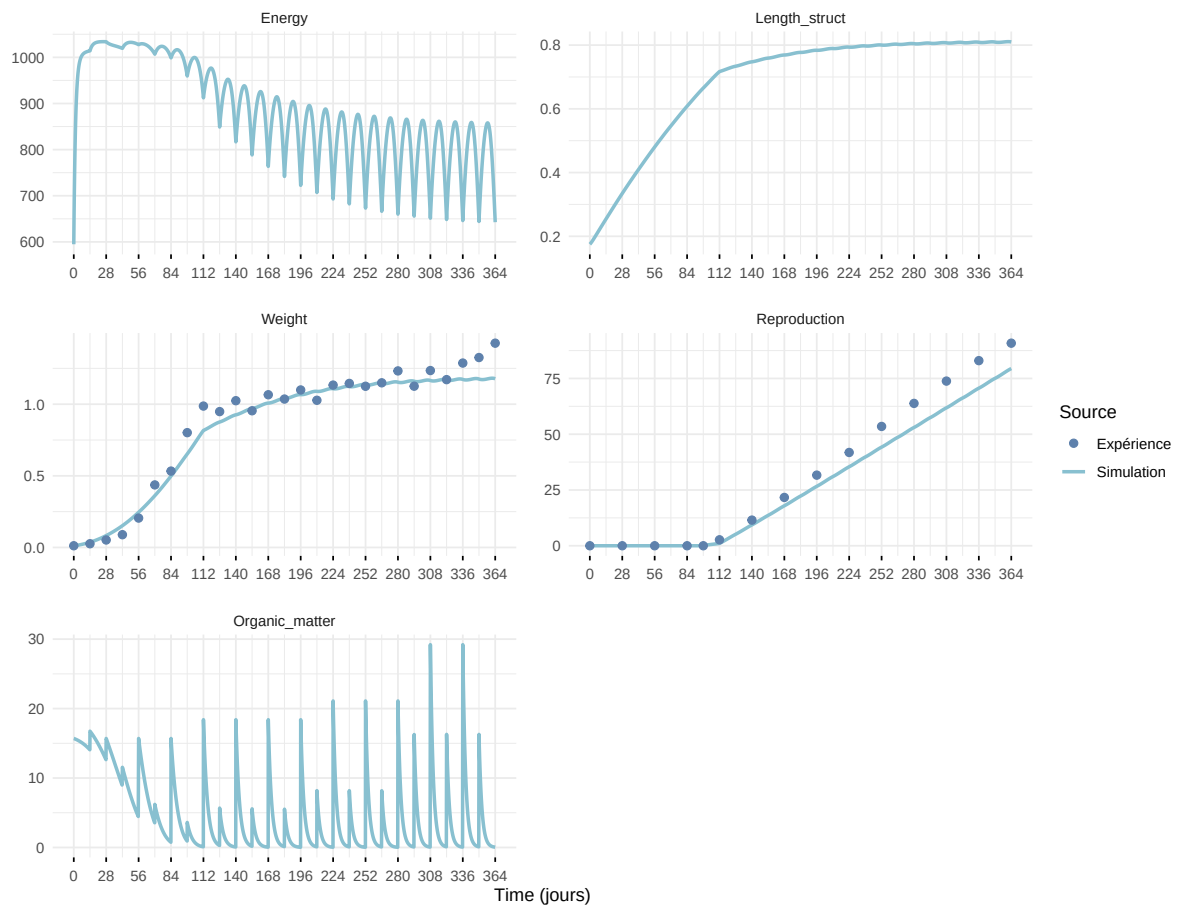
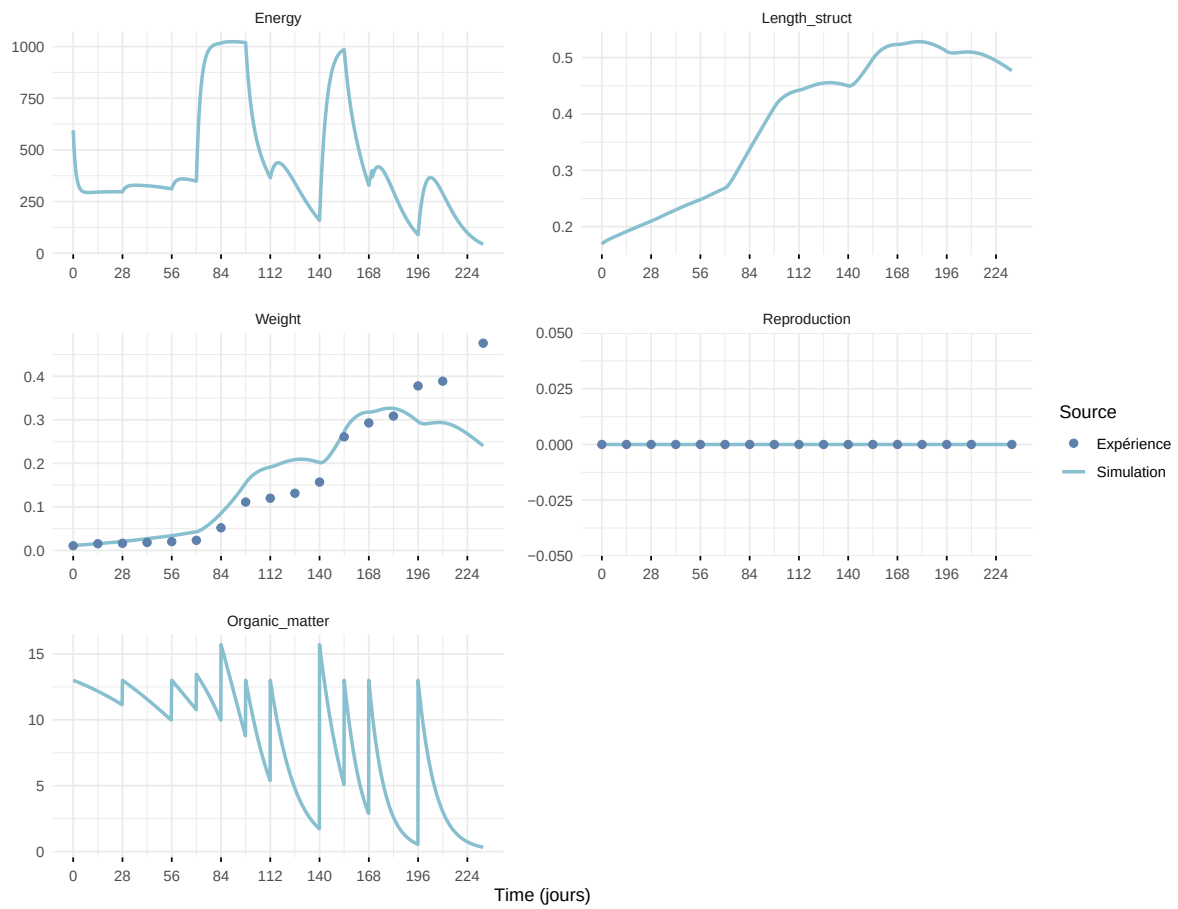


Figure 7: Simulations Weight and Reproduction.

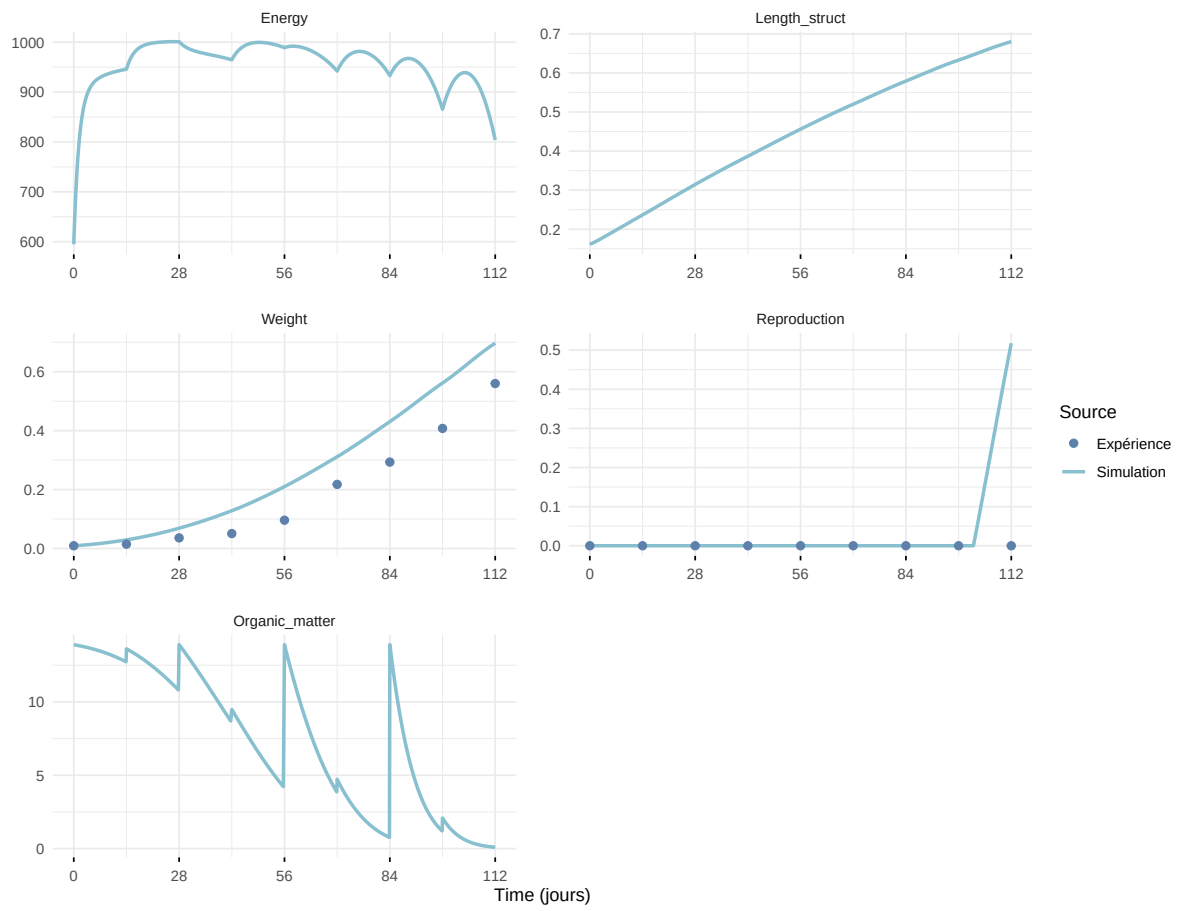
Expérience : Bart2019_XP1_2



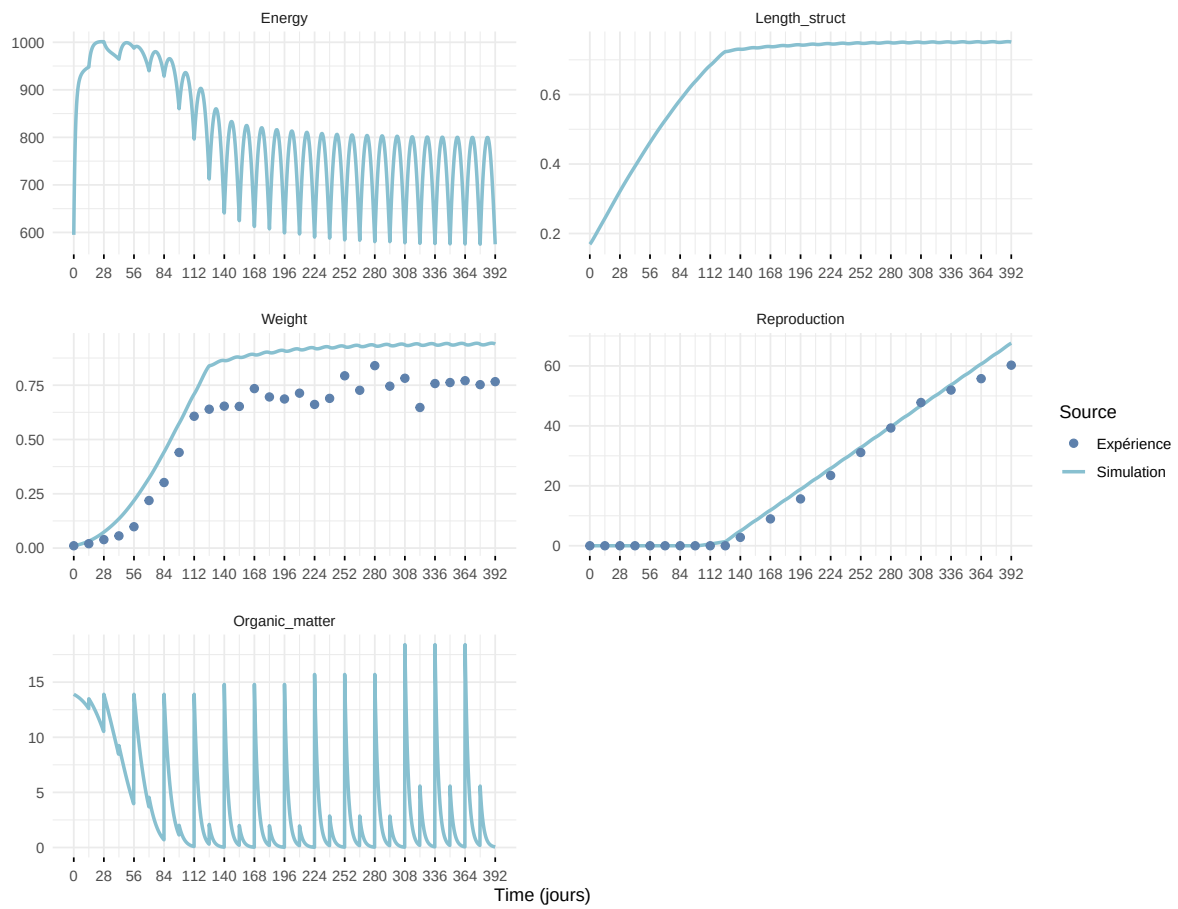
Expérience : Bart2019_XP3



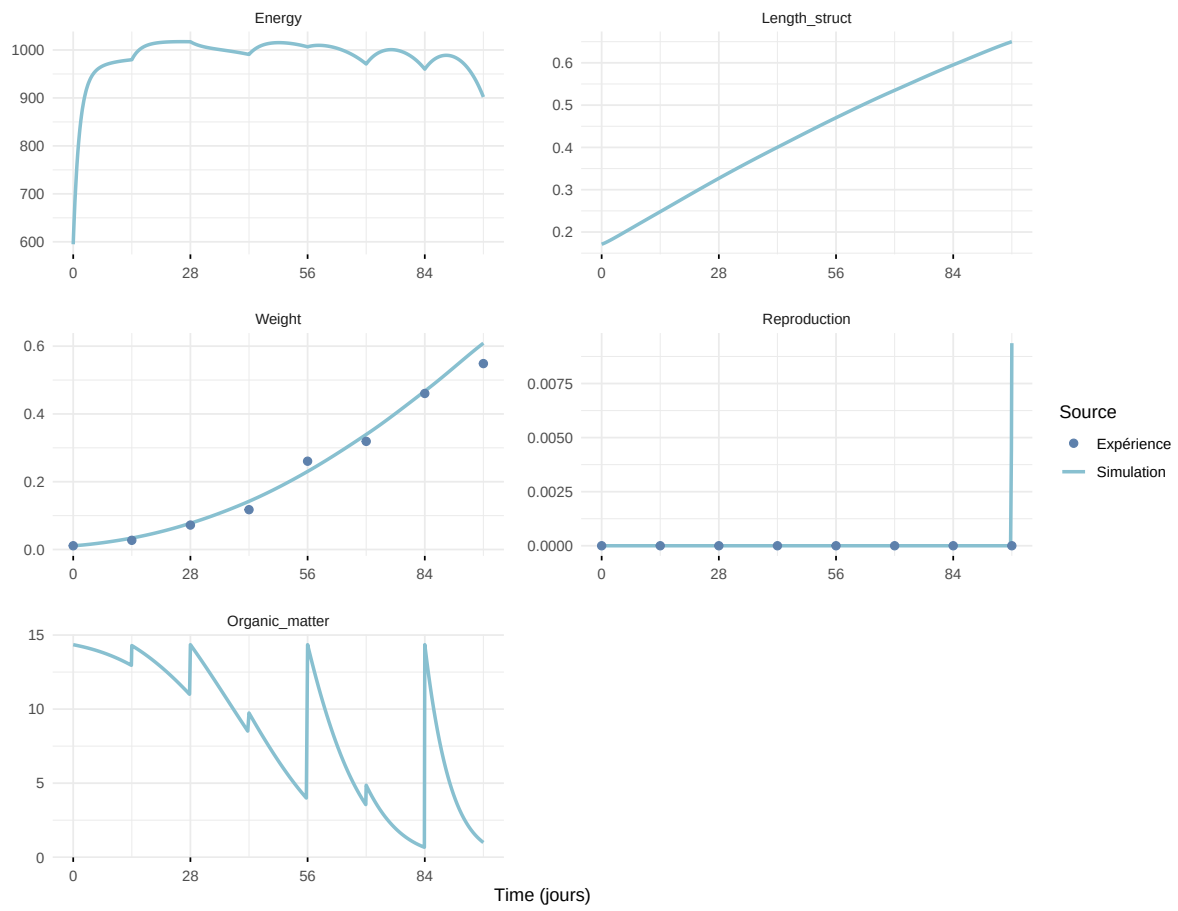
Expérience : Bart2019_XP4_1



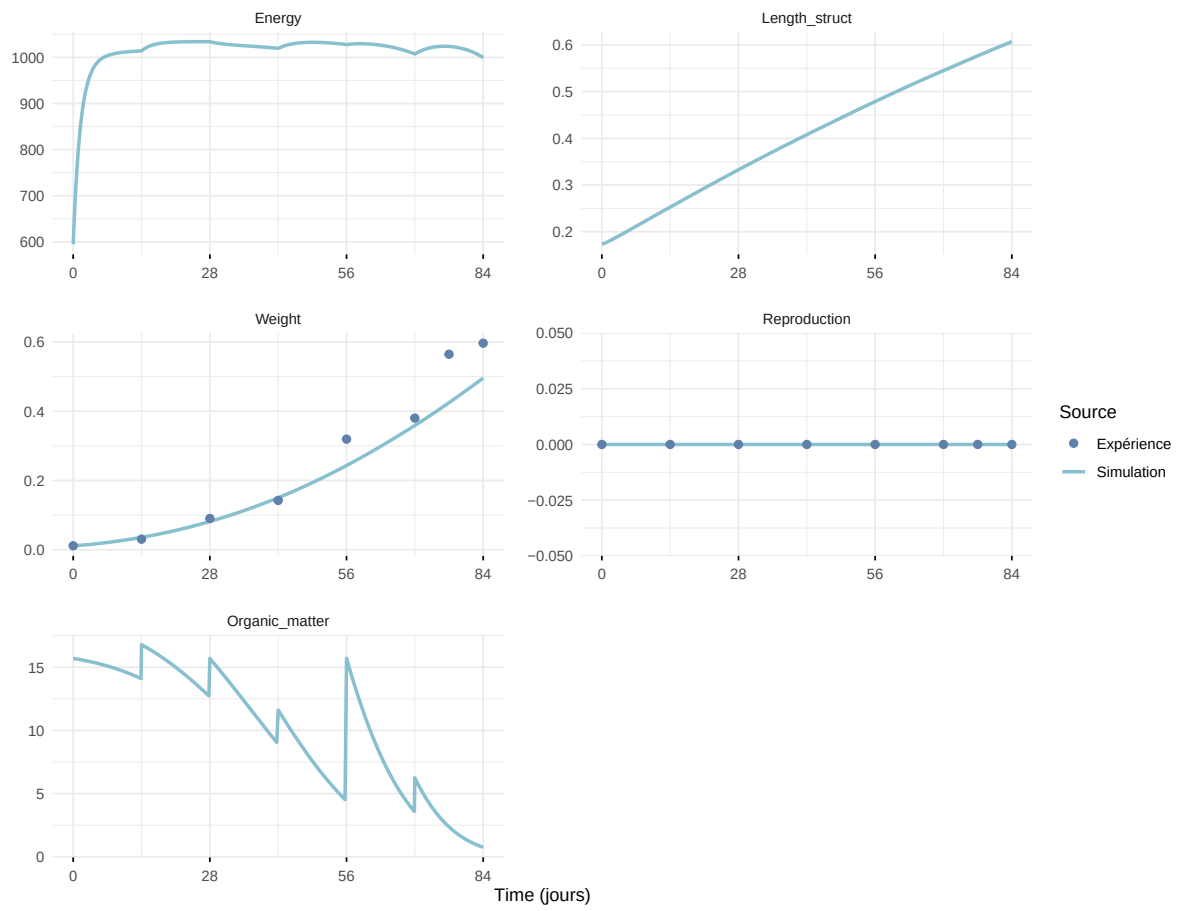
Expérience : Bart2019_XP4_2



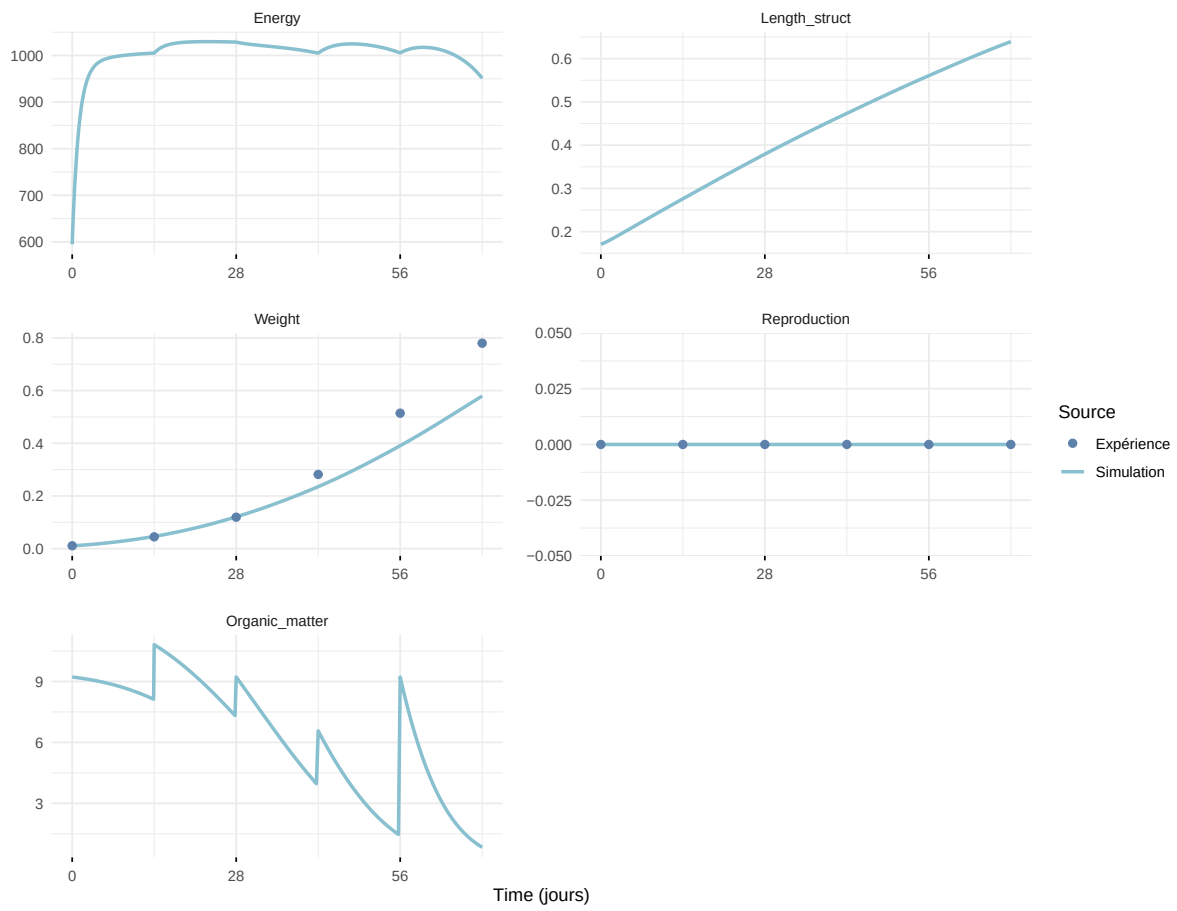
Expérience : Bart2019_XP5



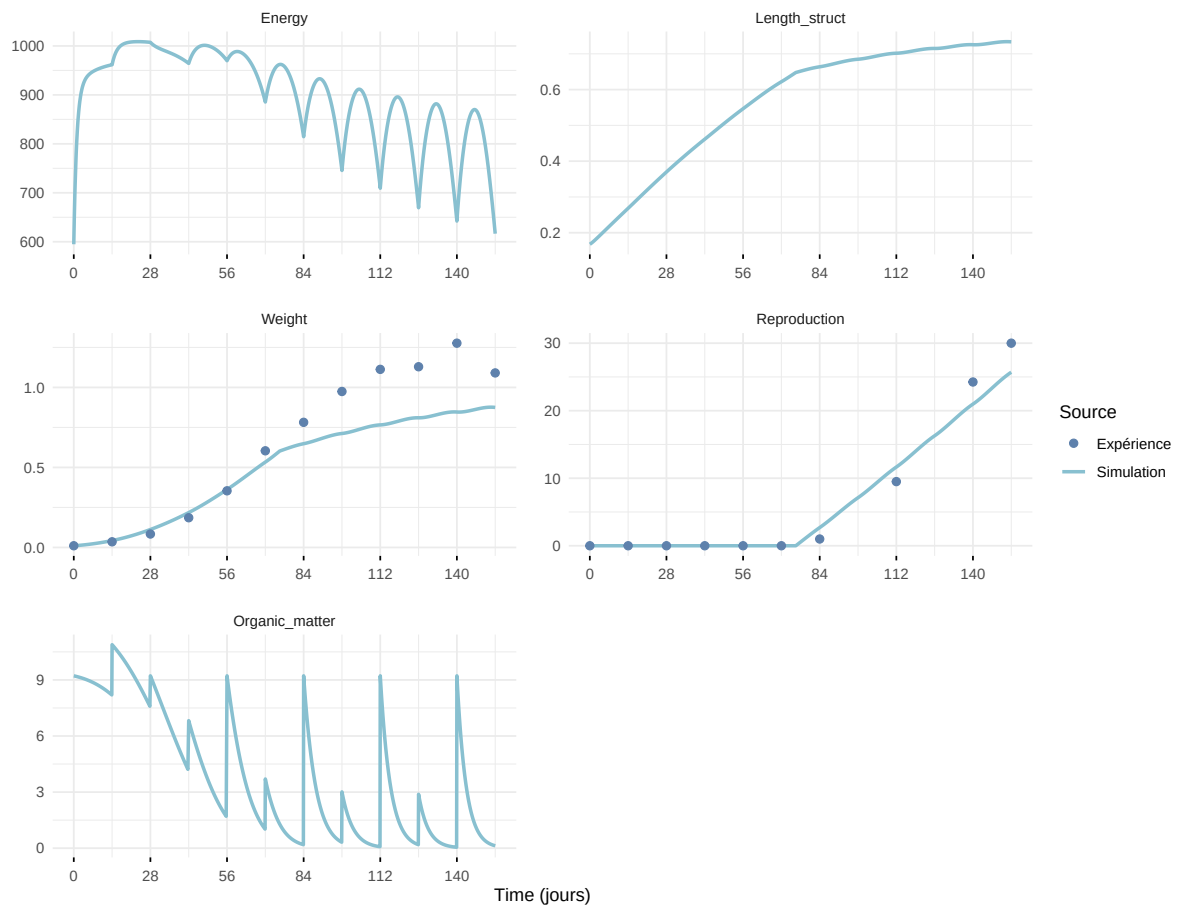
Expérience : Bart2020



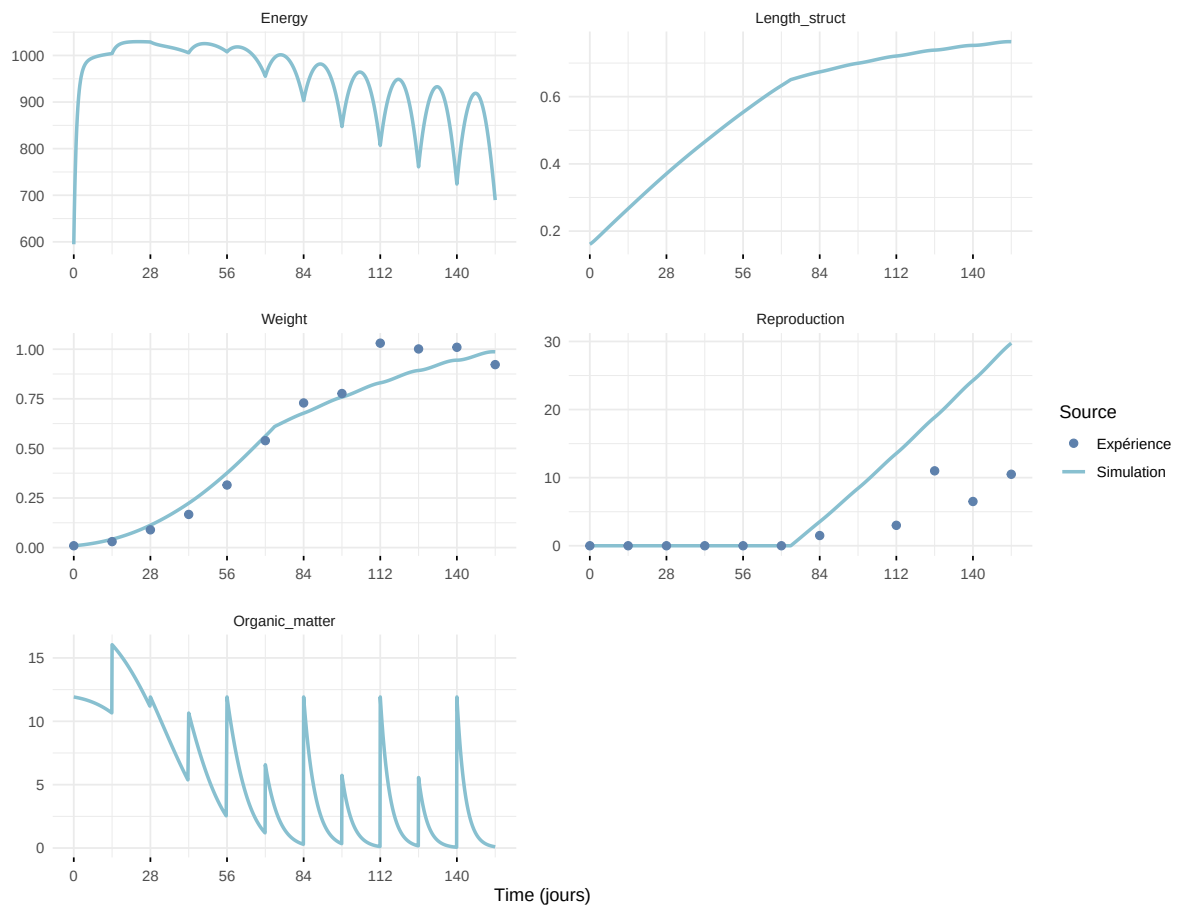
Expérience : Gollot2026_dens_D1



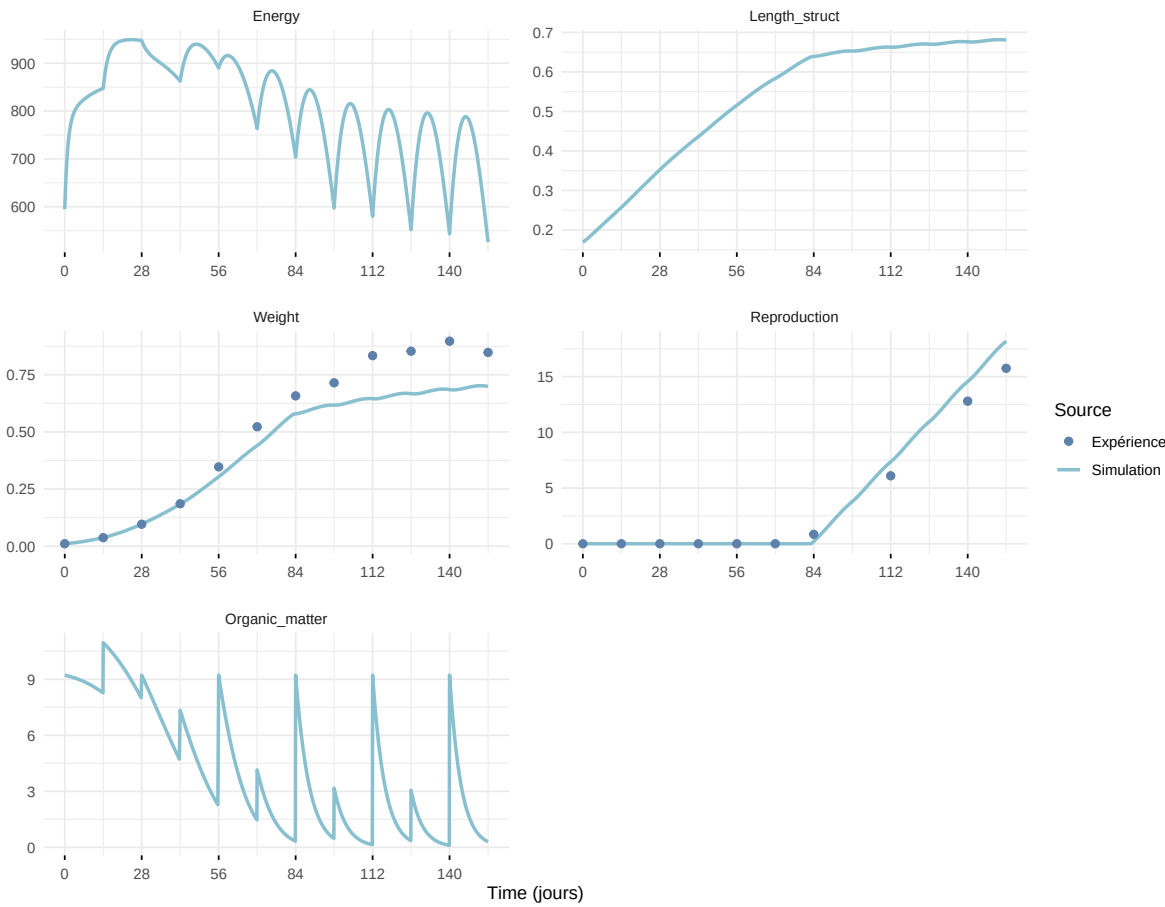
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

